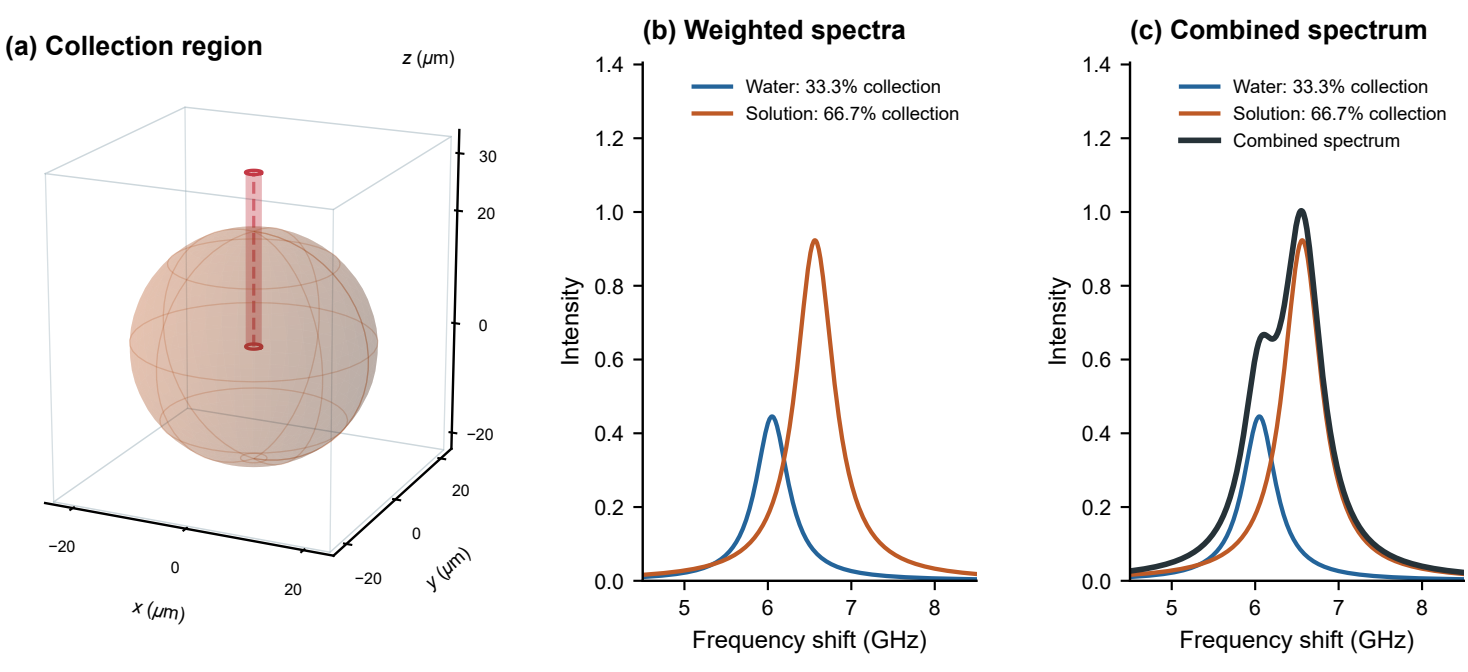

(a) Collection region
z (μm)
30
20
0
−20
20
0
−20
−20
0
20
x (μm)
y (μm)
(b) Weighted spectra
Water: 33.3% collection
Solution: 66.7% collection
Intensity
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
5
6
7
8
Frequency shift (GHz)
(c) Combined spectrum
Water: 33.3% collection
Solution: 66.7% collection
Combined spectrum
Intensity
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
5
6
7
8
Frequency shift (GHz)